

The University of Nottingham Ningbo China

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, SPRING SEMESTER 2023-2024

Professional Ethics in Computing (COMP3056 UNNC)

Time allowed: **TWO Hours**

Candidates may complete the front cover of their answer book and sign their desk card but must NOT write anything else until the start of the examination period is announced.

Answer all TWO questions.

All questions are worth 25 marks each, hence the total mark is 50.

(Each question is worth equal marks)

No calculators are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn your examination paper over until instructed to do so

ADDITIONAL MATERIAL: *British Computer Society Code of Conduct.*

INFORMATION FOR INVIGILATORS: Exam papers must be collected at the end of the exam.

i) The “Nothing to Hide” argument: Richard Posner believes privacy as a social good is greatly overrated. The “Nothing to Hide” argument claims that individuals who have done nothing wrong have no reason to be concerned about privacy or surveillance. AE3PEC-E1 AS23/24

Richard Posner supports a related view by treating privacy primarily as concealment and arguing that its value as a social good is often overstated.

Question 1 (25 marks)

This argument is often criticised for ignoring the broader roles of privacy in protecting autonomy, dignity, and individuals from misuse of information.

a) Answer the following questions about *Privacy*. lec 7

ii) Some argue that privacy is merely derived from other rights, such as implicit contracts or intellectual property.

(i) Explain the idea of privacy as concealment, according to Richard Posner. [3 marks]

However, Warren and Brandeis refute this by showing that cases like the publication of private letters cannot be adequately addressed by contract or intellectual property law. This suggests that privacy should be recognised as an independent right.

(ii) Give one argument in favour of privacy as concealment, and one against. [2 marks]

iii) Solove introduces a hierarchical taxonomy of privacy to move away from abstract definitions and instead focus on concrete privacy violations. By categorising violations into information collection, processing, dissemination, and invasion, the framework enables clearer identification and analysis of privacy harms. (iii) Explain the motivation for Solove’s Taxonomy of Privacy. [3 marks]

lec9-p8 b) i)

Hardware errors

Software errors

Computer solves the wrong problem

Misuse

Communication failure (in human interaction)

Malice

b) Answer the following questions about *Computer Failure and Trust*.

(i) List five different key reasons for computer failures. [5 marks]

Hint: Your answer should be a simple list with one line per item.

ii) According to Birsch, three criteria must be met for a person to bear moral responsibility:

1. “The person’s actions must have caused the harm, or been a significant causal factor

2. The person must have intended or willed the harm, or it must be a result of his/her negligence, carelessness, or recklessness

3. The person must be able to have known, or must know the consequences of the action, or must have deliberately remained ignorant of them

(ii) Give an example of when a programmer would be responsible for a software error with reference to Birsch’s 3 criteria. [4 marks]

For example, in the Therac-25 case, the programmers created or allowed software errors that directly caused lethal radiation overdoses.

They should have known that the system was safety-critical and that software failures could have severe consequences, especially given the removal of hardware safety interlocks. Their failure to implement adequate safeguards and testing therefore constitutes negligence, satisfying all three of Birsch’s criteria for moral responsibility.

c) Answer the following questions about Computing and Self.

(i) Which of these statements are true, and which are false? [4 marks]

A. **Griefing** is when one or more people spoil the enjoyment or activity of other online users for their own pleasure, especially in the context of online games. True

B. **Trolling** is when a person deliberately sets up an ugly persona on the internet. False

C. A **cyberbully** is someone who uses the internet to harass a particular target, often using fake identities or public web sites to enable the harassment. True

D. **Self-concept** is the collection of ideas and feelings that we have about ourselves. True

(ii) The internet encourages people to develop sociopathic behaviour.

Give reasons that support this statement. [4 marks]

The internet can encourage behaviour that appears sociopathic because computing systems mediate human interaction in ways that weaken traditional social constraints.

Online interactions often lack immediate social feedback and clear responsibility, making individuals less aware of the impact of their actions on others.

AE3PEC

Turn over

In addition, system design can reduce accountability and amplify harmful behaviour by allowing users to act without facing direct consequences.

Question 2 (25 marks)

Please read the following scenario carefully and answer the questions.
All agents and events in this scenario are fictitious.

The Horseshire Health Service (HHS) is seeking to procure a heart disease alert system from Zing Software Development Company. Zing have used state-of-the-art machine learning techniques to develop the system. A medical doctor can enter a patient's personal data and medical measurements and the system is able to give an alert for risk of heart disease. If the system indicates high risk, then the medical doctor can investigate further and prescribe appropriate lifestyle changes and/or medication to prevent heart disease.

Erin works as a programmer and software tester for HHS and she has been working on the testing team for the new heart disease alert system. Whenever she identifies a failed test, the correct procedure is to raise a fault report form which is then received by both HHS and Zing and needs to be formally resolved, typically with a software fix.

The tests go well until, when she is almost complete, she performs a test based on historical data, at which point she determines that the system is biased for an ethnic minority population that represents 2% of the Horseshire population. In particular, the system will significantly underestimate risk of heart disease within this ethnic minority. Erin is concerned by her discovery of this bug and is not sure what to do. Because she is a programmer and knows the system, she is aware that if the bias is genuine, it will be hard to fix. Dealing with the bug could be costly and she estimates would delay the release of the system by several weeks. With this information, these are some options that she considers:-

- (i) Formally complete and raise a fault report form.
- (ii) She may have made a mistake running the test, so she should keep quiet. She reasons that if it is a genuine problem, another tester will also find it and raise it.
- (iii) Claude, one of the lead developers at Zing, has become her friend over the course of the project. To save him from embarrassment, she should talk to him informally first, rather than raising a formal fault report. She reasons that Claude may be able to fix the problem quickly and quietly, causing less trouble.

a) Agents and relationships:

HHS is the purchasing and deploying organisation responsible for commissioning the system.

Zing is the software development company supplying the system.

AE3PEC-E1 AS23/24

Erin is a programmer and tester employed by HHS.

Claude is the lead developer at Zing and also Erin's friend, which introduces a potential conflict of interest.

Hamley is Erin's manager and represents the relevant authority within HHS.

- (iv) She should discuss the bug with her manager Hamley first to decide on the correct course of action, given the potential negative consequences of the bug.

Medical doctors are the end users of the system, relying on it for decision support.

Patients, including the affected 2% ethnic minority, are the subjects impacted by the system's decisions.

a) Identify all the agents and their relationships in this scenario. [5 marks]

d) Erin should raise a formal fault report and escalate the issue to senior management in order to ensure that the risks are properly assessed and addressed.

If internal reporting mechanisms fail and the system poses serious and foreseeable harm to patients, whistleblowing may be ethically justified as a last resort.

b) Discuss Erin's options (i) to (iv) in relation to virtue theory, act

utilitarianism, deontological ethics and the ethics of care. [8 marks]

Erin should also contribute to resolving the bug through appropriate organisational procedures and recommend delaying deployment until the issue is fixed.

In addition, the system should be used strictly as a decision-support tool.

c) Identify one item from the BCS Code of Conduct that could be breached for each of the options (i) to (iv), and explain why. [8 marks]

Medical doctors should remain in the loop and exercise professional judgement rather than relying on the software alone when diagnosing patients.

Suppose Erin chooses option (iv) and talks to Hamley. He tells Erin to take no further action on the grounds that the bug only affects a small number of people, if any at all, and so long as it is working well for the majority of people, that is the most important thing.

d) Discuss Erin's further options, with professional and ethical

considerations, following Hamley's instruction. [4 marks]

(i) From a virtue ethics perspective, it demonstrates professional integrity, honesty, and courage.

Act utilitarianism supports this option because reporting the fault helps prevent potentially serious harm to patients, maximising overall wellbeing.

From a deontological perspective, Erin fulfils her professional duty to avoid harm and to act responsibly.

Ethics of care also supports this option, as it prioritises the welfare of vulnerable groups, including the affected ethnic minority.

[END OF EXAM PAPER]

(ii) Keeping quiet and hoping others discover the problem is ethically unjustifiable.

It reflects a lack of professional integrity under virtue ethics and risks serious harm to patients, which act utilitarianism would reject.

Deontologically, knowingly allowing harm to occur violates Erin's duty as a computing professional, and ethics of care condemns the neglect of vulnerable groups.

(iii) Informally telling Claude presents mixed ethical considerations.

While it may appear caring toward a friend, it risks bypassing formal procedures and introduces a conflict of interest.

From a deontological perspective, this option is problematic because it undermines professional responsibility and accountability, and from a utilitarian view it risks greater harm if the informal approach fails.

(iv) Discussing the issue with Hamley is ethically acceptable and preferable to informal action.

It reflects professional integrity under virtue ethics, supports act utilitarianism by enabling organisational risk management, satisfies deontological duties to follow appropriate reporting structures, and aligns with ethics of care by seeking to protect patients through institutional action.

Overall, options (i) and (iv) are ethically preferable, with (i) being the strongest, while (ii) is clearly unethical and (iii) is ethically risky.

c) (i) Formal fault report

This option does not constitute a breach of the BCS Code of Conduct.

On the contrary, it is consistent with Public Interest, as it protects public health and wellbeing, and with Professional Competence and Integrity, as Erin reports known defects honestly and responsibly.

(ii) Keep quiet

This option breaches the Public Interest principle by failing to protect patients from foreseeable harm caused by a biased medical system.

It also violates Professional Competence and Integrity, as relying on others to discover the problem represents negligent inaction rather than due diligence.

(iii) Talk informally to Claude

This option risks breaching the Duty to Relevant Authority, as it bypasses formal fault-reporting procedures and introduces a conflict of interest due to personal friendship.

It may also undermine Professional Integrity by prioritising informal relationships over transparent and accountable processes.

(iv) Talk to HamleyAE3PEC

This option does not constitute a breach of the Code.

Instead, it satisfies the Duty to Relevant Authority, as Erin follows appropriate organisational reporting structures while exercising professional judgement.